

Course “Theoretical Ecotoxicology: Concepts and applications”

Introduction to Ecological Modelling II

Hanh H. Nguyen

University of Duisburg-Essen
& Research Center One Health Ruhr

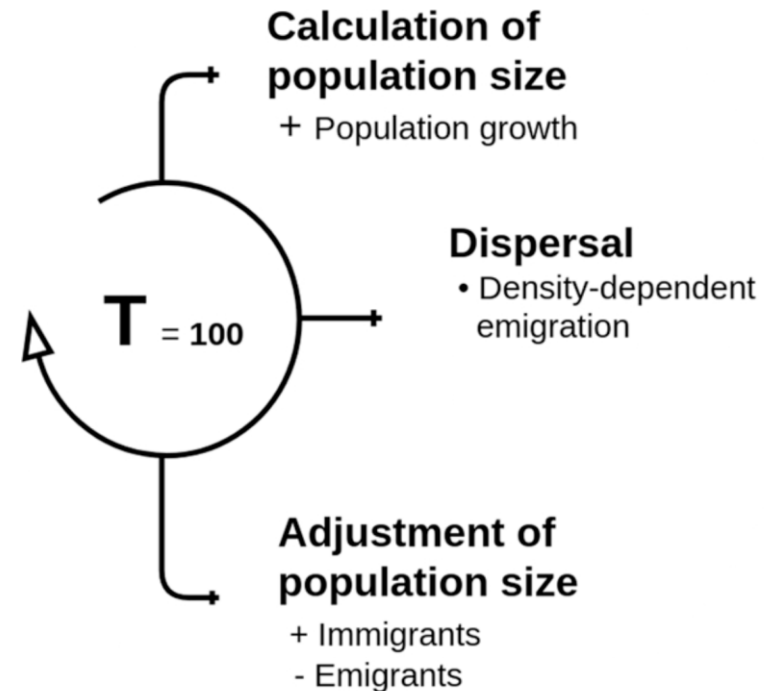
Learning topics – Ecological modelling II

I. Non-spatial approaches - Population dynamics:

- Population growth model
- Species interaction models

II. Spatio-temporal dynamics:

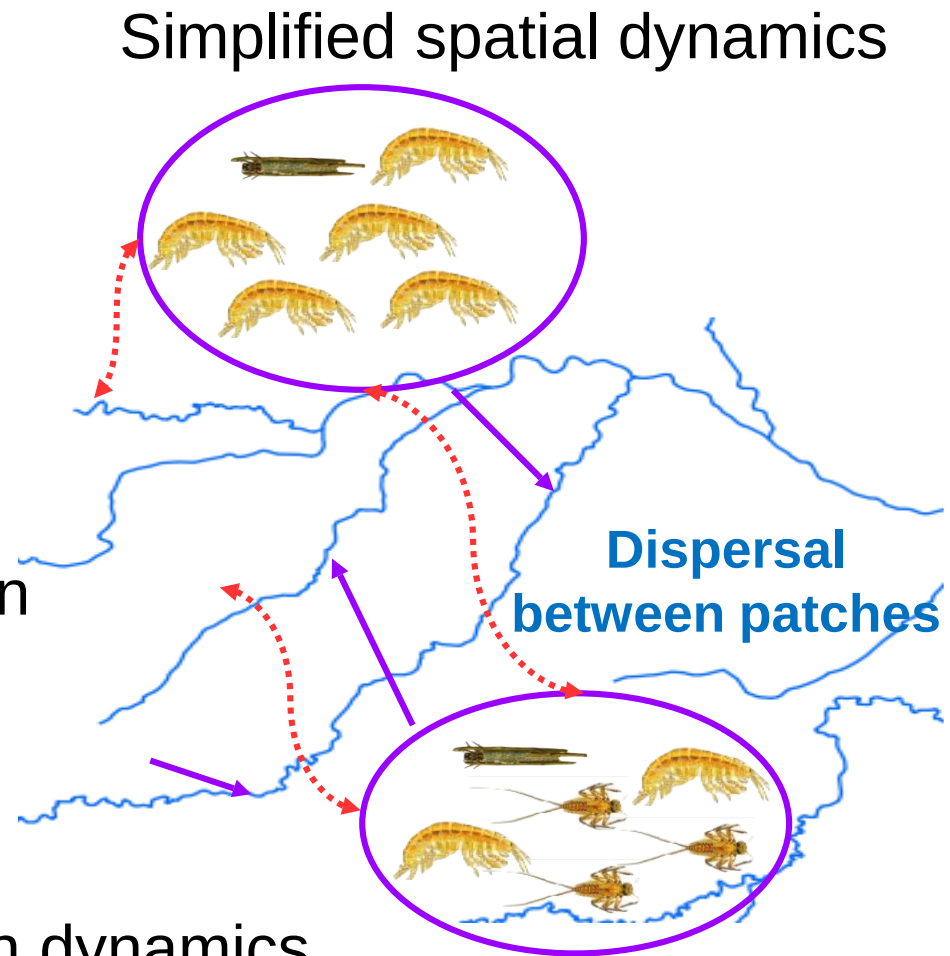
- Population in space
- Metapopulation models
- Metacommunity models



Introduction

- Empirical studies often focus on species dynamics within disconnected patches
- Example: population growth models (closed systems)
- Real population dynamics can be more complex

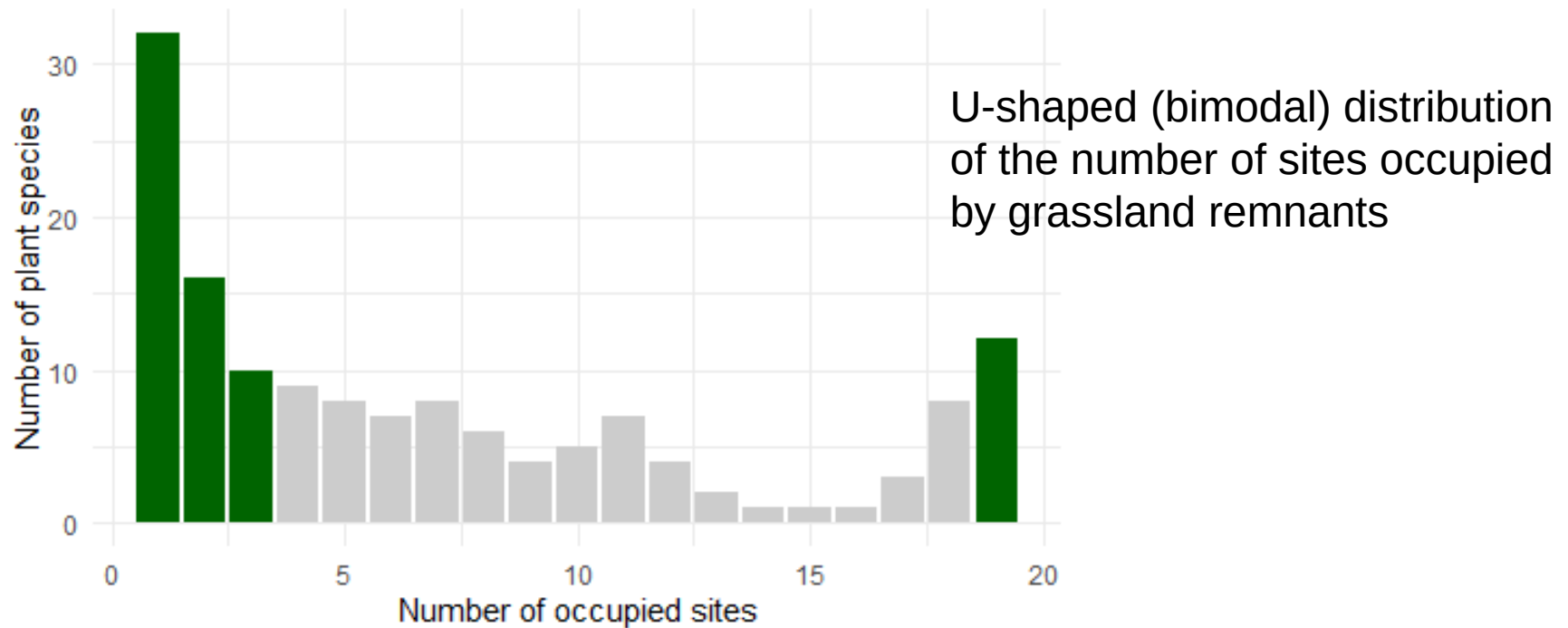
→ **Space** influence population dynamics



Populations in space

In a spatial context:

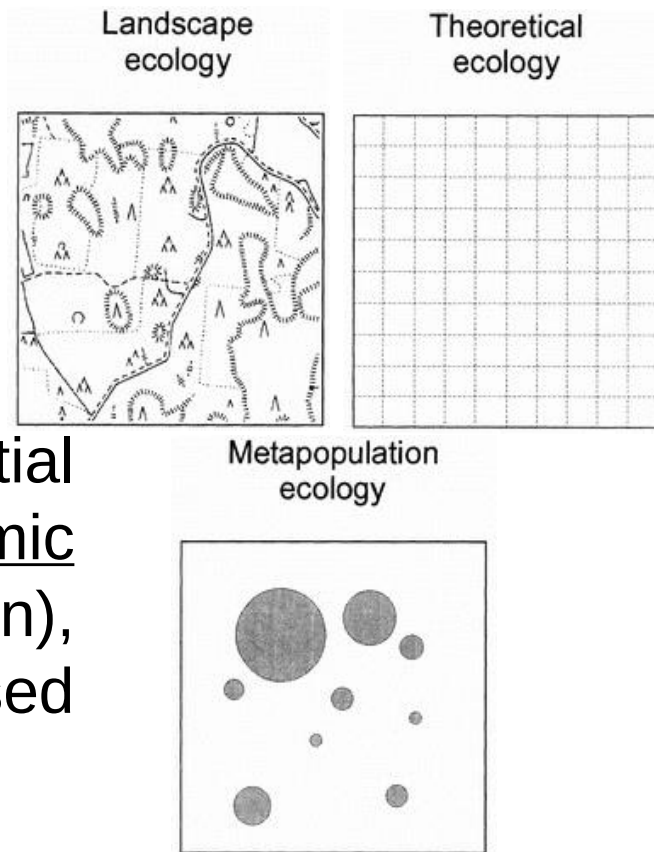
- Some species have narrow distributions
- Other species can occupy large areas



→ Assumption: Effects of emigration and immigration

Approaches to spatial ecology

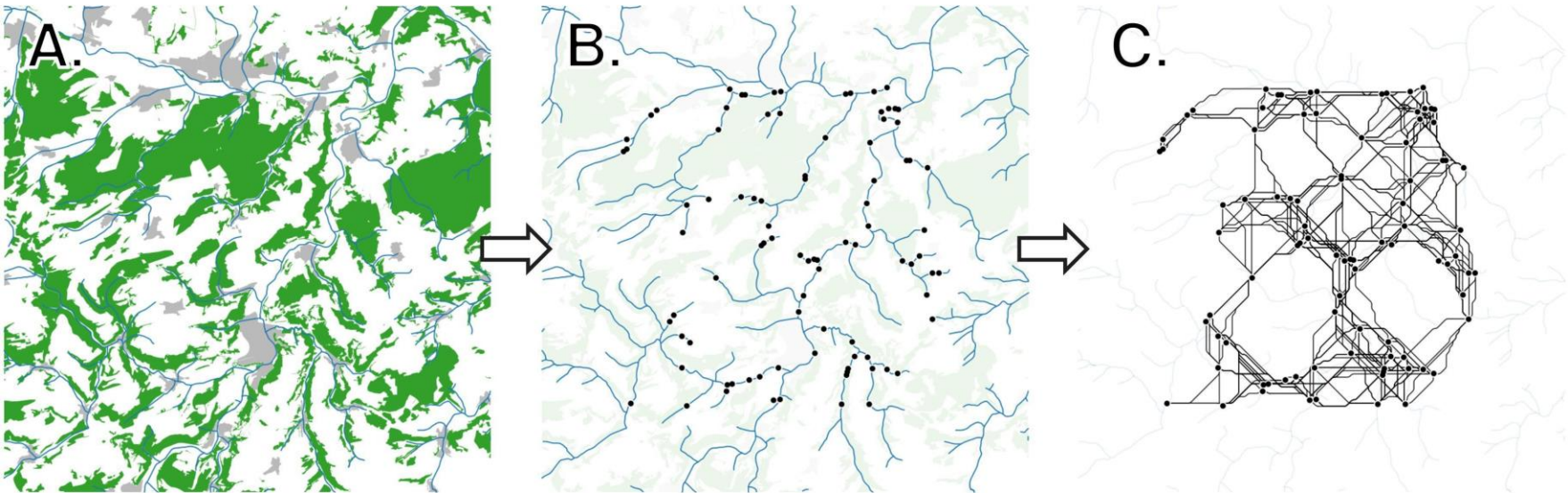
- **Landscape ecology:** focuses on landscape structure and heterogeneity, and effects of human-induced environmental changes on biodiversity and ecosystem functioning.
- **Theoretical ecology:** prioritize spatial organization of population dynamic processes (e.g., dispersal, competition), with assumed homogeneous or discretised space (e.g., grids).



- **Metapopulation ecology:** focuses on patch-occupancy population dynamics and connectivity via species immigration/emigration, with landscapes treated as a network of connected patches.

Metapopulation

- Metapopulations: are populations of populations (Levins, 1969)
- Populations of spatially connected discrete habitat patches interact via dispersal
- Metapopulation models address species persistence in spatially heterogeneous environment, important for conservation.



Types of metapopulation models

(1) Patch occupancy models:

- Define whether a location is occupied by a species using presence–absence data
- Require information on occupancy of neighbouring connected locations

(2) Metapopulation models (with explicit movement):

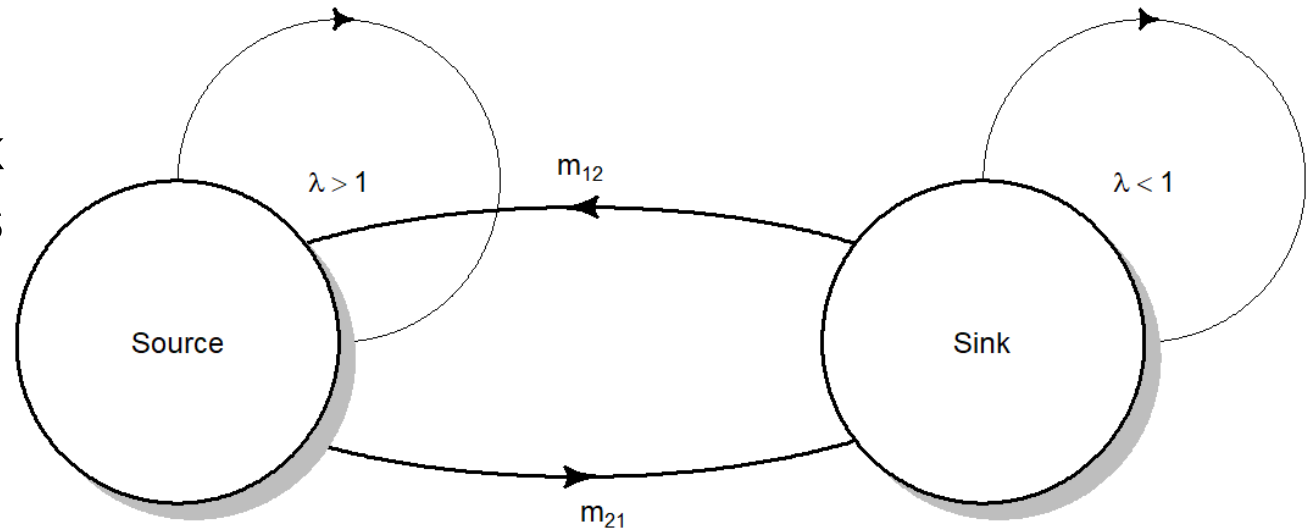
- Movement between locations is described explicitly
- More realistic simulation of complex dynamics (e.g., rescue effects and local extinctions), but require more detailed data and parameterization

Source – sink dynamics

- Assumption: Spatially separated sub-populations occupy distinct patches, each with intrinsic B-I-D-E dynamics

Birth – Immigration – Death – Emigration

Concept of
source-sink
populations



λ = population growth rate

m = migration of individuals between patches ($m_{21} > m_{12}$)

- Source: $\lambda > 1$ and emigration $>$ immigration
- Sink: $\lambda < 1$ and immigration $>$ emigration

Metapopulation models

- Classic metapopulation model (Levins, 1969)

$$\frac{dp}{dt} = c_i p(1 - p) - ep$$

- p = proportion of patches occupied at a given time
- c_i = colonization rate
- e = extinction rate of local populations

- Main assumptions

- Patches are the same size and equally accessible
- Local patch population dynamics are ignored
- Each habitat patch is either occupied (1) or unoccupied (0)

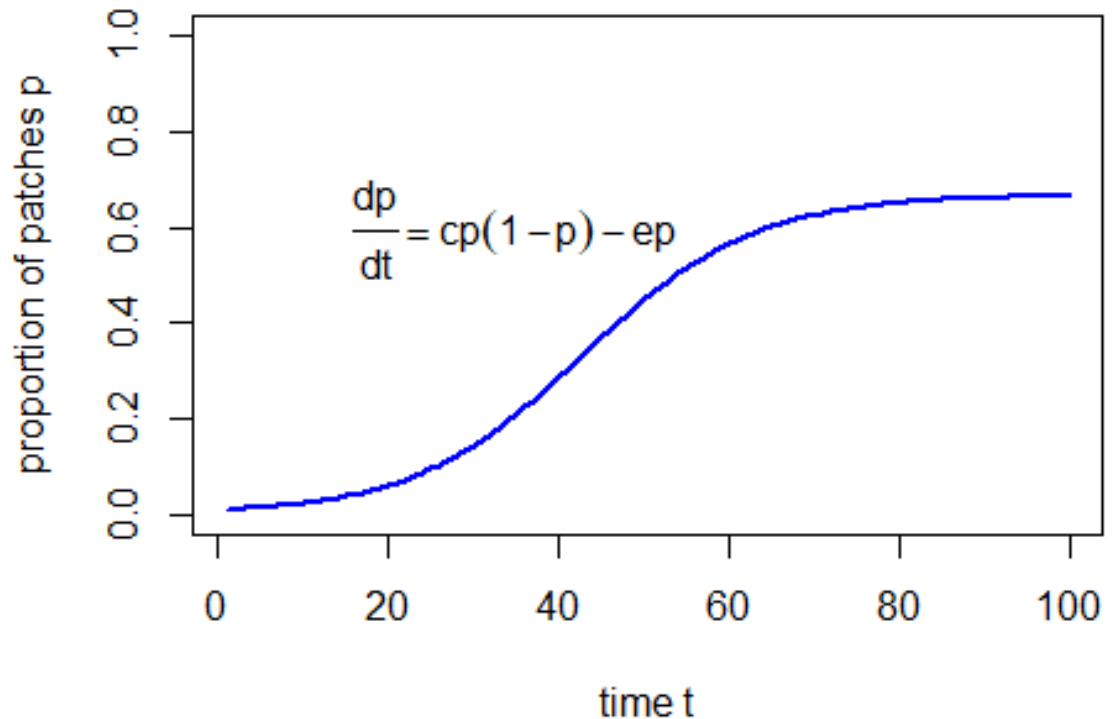
Metapopulation models - Equilibrium

• Equilibrium: $\frac{dp}{dt} = 0$

$$\Rightarrow c_i p(1 - p) - e p = c_i p - c_i p^2 - e p = 0$$

$$\Rightarrow p^* = \frac{c_i - e}{c_i} = 1 - \frac{e}{c_i}$$

• Extinction: if $e > c_i$



Exercise 1 - Levins metapopulation model

Q1.1. Implement the Levins metapopulation model in R.

Define model equilibrium.

Model function:

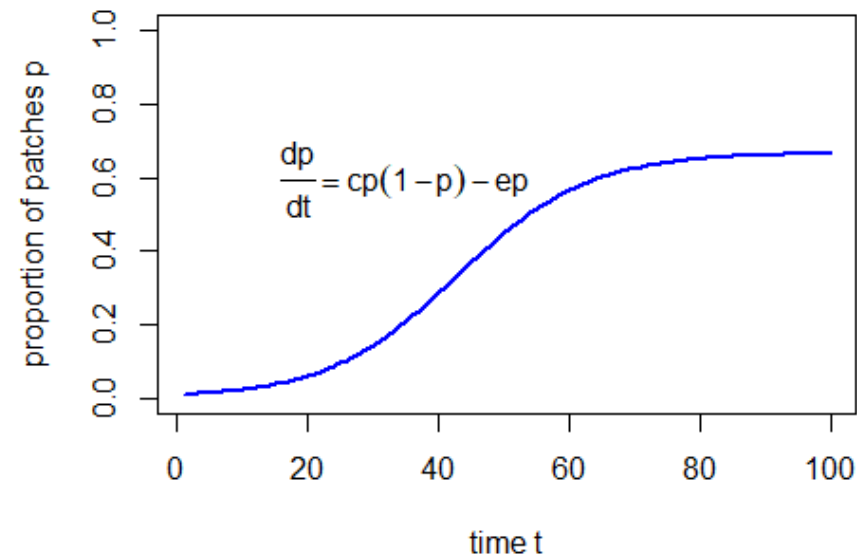
$$dp_dt = ci * p * (1 - p) - e * p$$

Model parameters:

```
times <- seq(from = 1, to = 100, by = 1)
```

```
parms <- c(ci=0.15, e=0.05)
```

```
init <- 0.01
```



Q1.2. Compare models – what are relations of (r and K) in

Verhulst population growth model and (ci and e) in Levins model?

$$\text{Verhulst model: } \frac{dN}{dt} = rN * \left(1 - \frac{N}{K}\right)$$

Exercise 1 - Levins metapopulation model

Q2. Compare model functions – what are relations of (r and K) in Verhulst population growth model and (ci and e) in Levins model?

$$\text{Levins: } \frac{dp}{dt} = c_i p(1 - p) - ep$$

$$\text{Verhulst: } \frac{dN}{dt} = rN * (1 - \frac{N}{K})$$

$$r = ci - e$$

$$K = 1 - \frac{e}{ci}$$



$$K = 1 - \frac{e}{ci} = \frac{ci - e}{ci} = \frac{r}{ci} \Rightarrow ci = \frac{r}{K}$$

$$\begin{aligned} \frac{dp}{dt} &= c_i p(1 - p) - ep = (ci - e) * p - ci * p * p \\ &= r * p - \frac{r}{K} * p * p = rp * (1 - \frac{p}{K}) \end{aligned}$$

$$\frac{dp}{dt} = rp * (1 - \frac{p}{K})$$

Metapopulation models – Propagule rain

$$\text{Levins: } \frac{dp}{dt} = c_i p (1 - p) - ep$$



- Including external colonisation via dispersal (Gotelli, 1991):

$$\frac{dp}{dt} = c_e (1 - p) - ep$$

c_e = external dispersal rate

- The equilibria depends on external (c_e) colonization
- External colonization reduces risks of local extinction and overall metapopulation decline

Metapopulation – Core satellite model

$$\text{Levins: } \frac{dp}{dt} = c_i p(1 - p) - ep$$



- Core satellite model (Hanski, 1982):

$$\frac{dp}{dt} = c_i p(1 - p) - ep(1 - p)$$

- Extinction rate decreases with an increase in p .
- 'Rescue effect': As $p \rightarrow 1$, dispersing propagules land on all sites, including sites on the verge of extinction.

Exercise 2 – Compare three metapopulation models

Q2.1. Implement three model types in R

(Classical model) Levins, 1969: $\frac{dp}{dt} = c_i p (1 - p) - e p$

(Propagule rain) Gotelli, 1991: $\frac{dp}{dt} = c_e (1 - p) - e p$

(Core-satellite) Hanski, 1982: $\frac{dp}{dt} = c_i p (1 - p) - e p (1 - p)$

Q2.2. Plot them jointly in one plot for a set of parameters:

```
times <- seq(from = 1, to = 100, by = 1)
```

```
init <- 0.01
```

```
e <- 0.05
```

```
ci <- ce <- c <- 0.15
```

```
parms <- c(c=0.15, e=0.05)
```

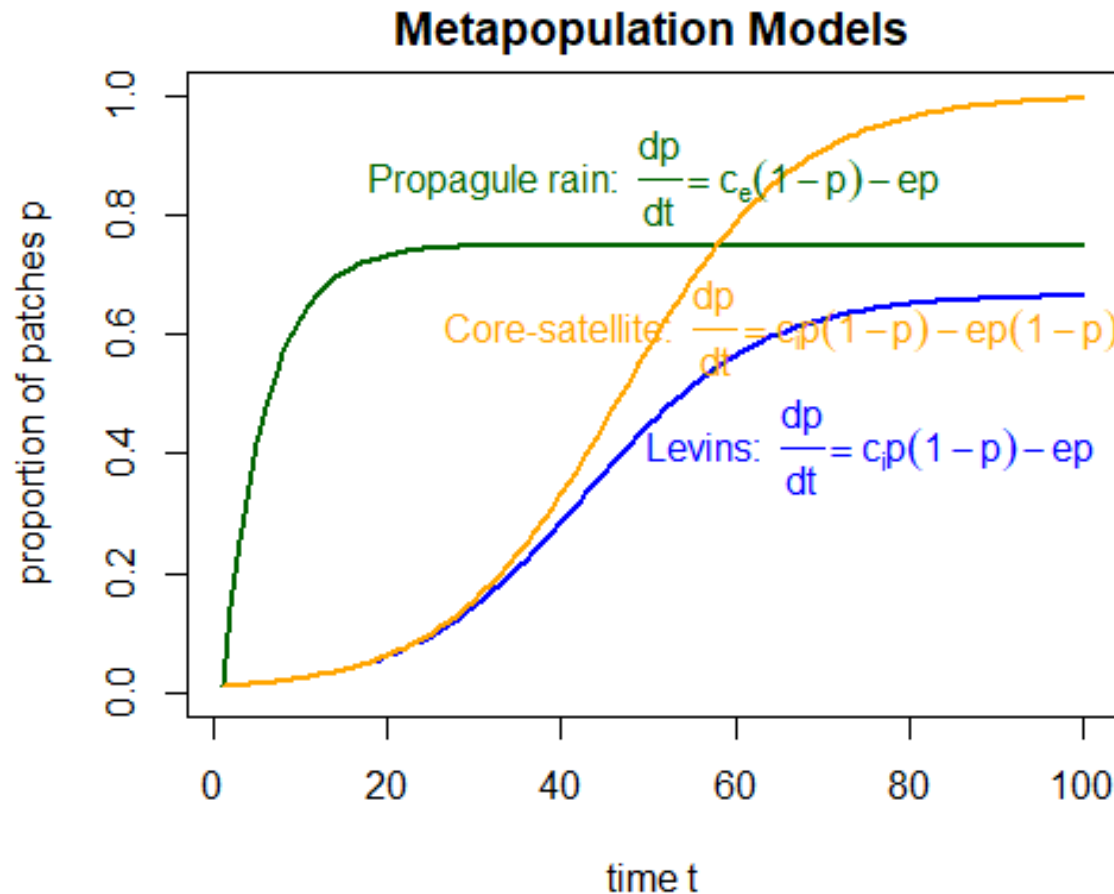
Q2.3. Determine equilibrium conditions p of each model ($\frac{dp}{dt} = 0$).

Exercise 2 – Compare three metapopulation models

(Classical model) Levins, 1969: $\frac{dp}{dt} = c_i p (1 - p) - e p$

(Propagule rain) Gotelli, 1991: $\frac{dp}{dt} = c_e (1 - p) - e p$

(Core-satellite) Hanski, 1982: $\frac{dp}{dt} = c_i p (1 - p) - e p (1 - p)$

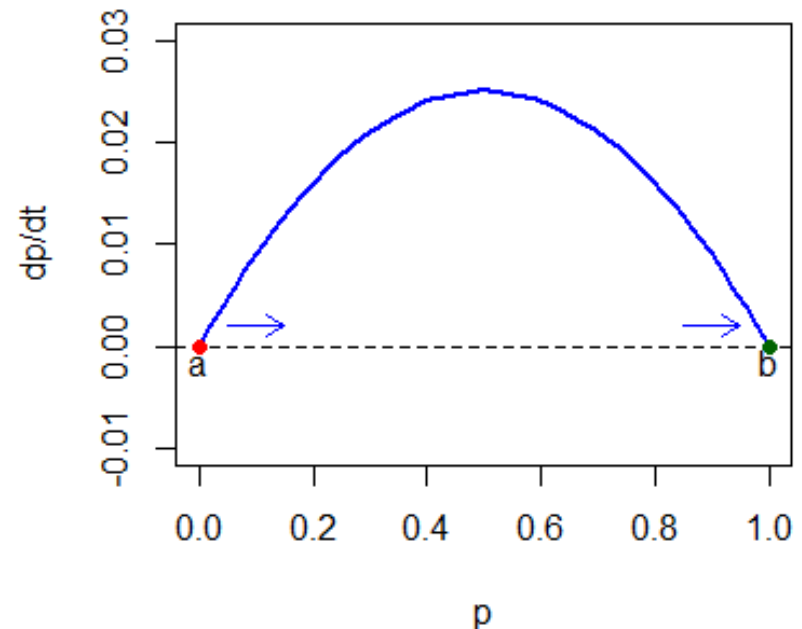


Stability of equilibria

- Core-satellite model (Hanski, 1982): $\frac{dp}{dt} = c_i p(1-p) - ep(1-p)$
- Equilibrium conditions: $\frac{dp}{dt} = 0$
- We define partial derivative $\frac{\partial \dot{p}}{\partial p}$ (slope of time derivative $\dot{p}$) with respect to p is:

$$\begin{aligned}\frac{dp}{dt} = \dot{p} &\Rightarrow \frac{\partial \dot{p}}{\partial p} = \frac{\partial (c_i - e)p(1-p)}{\partial p} \\ &= (c_i - e)(1 - 2p)\end{aligned}$$

Phase diagram: Hanski model



Stability of equilibria

$$\text{Partial derivative: } \frac{\partial \dot{p}}{\partial p} = \frac{\partial (c_i - e)p(1-p)}{\partial p} = (c_i - e)(1 - 2p)$$

- $c_i > e$:

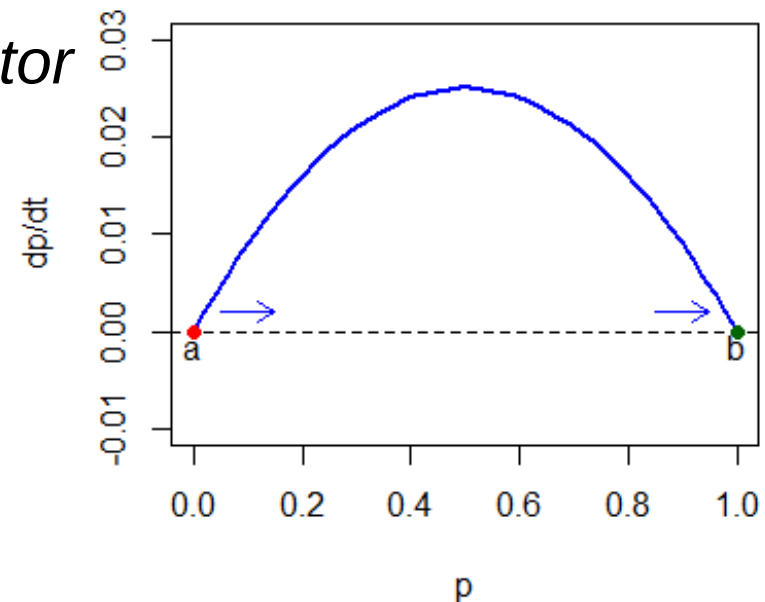
$$\text{At } p^* = 1 \Rightarrow \frac{\partial \dot{p}}{\partial p} < 0$$

$\Rightarrow$ stable equilibrium or stable attractor

$$\text{At } p^* = 0 \Rightarrow \frac{\partial \dot{p}}{\partial p} > 0$$

$\Rightarrow$ unstable equilibrium or repellor

Phase diagram: Hanski model



Stability of equilibria

$$\text{Partial derivative: } \frac{\partial \dot{p}}{\partial p} = \frac{\partial (c_i - e)p(1-p)}{\partial p} = (c_i - e)(1 - 2p)$$

- $c_i < e$:

At $p^* = 1 \Rightarrow \frac{\partial \dot{p}}{\partial p} > 0 \Rightarrow$ *unstable equilibrium or repellor*

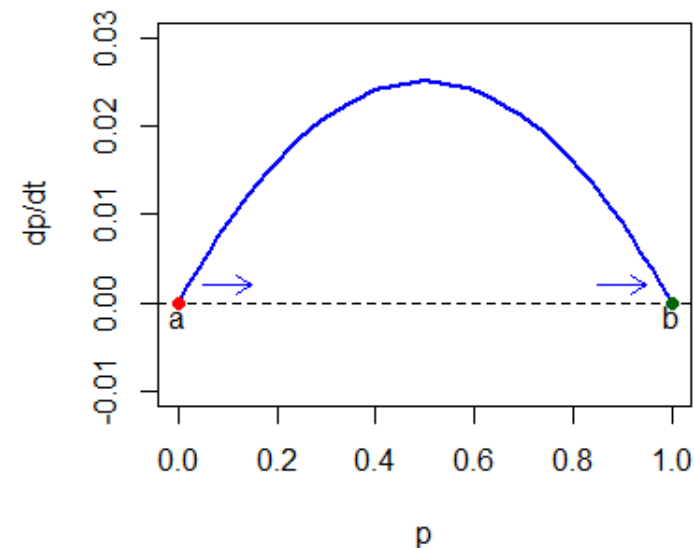
At $p^* = 0 \Rightarrow \frac{\partial \dot{p}}{\partial p} < 0 \Rightarrow$ *stable equilibrium or stable attractor*

- $c_i = e$:

$$\text{Both } \frac{\partial p}{dt} = \dot{p} = 0 \text{ and } \frac{\partial \dot{p}}{\partial p} = 0$$

$\Rightarrow$ *neutral equilibrium*

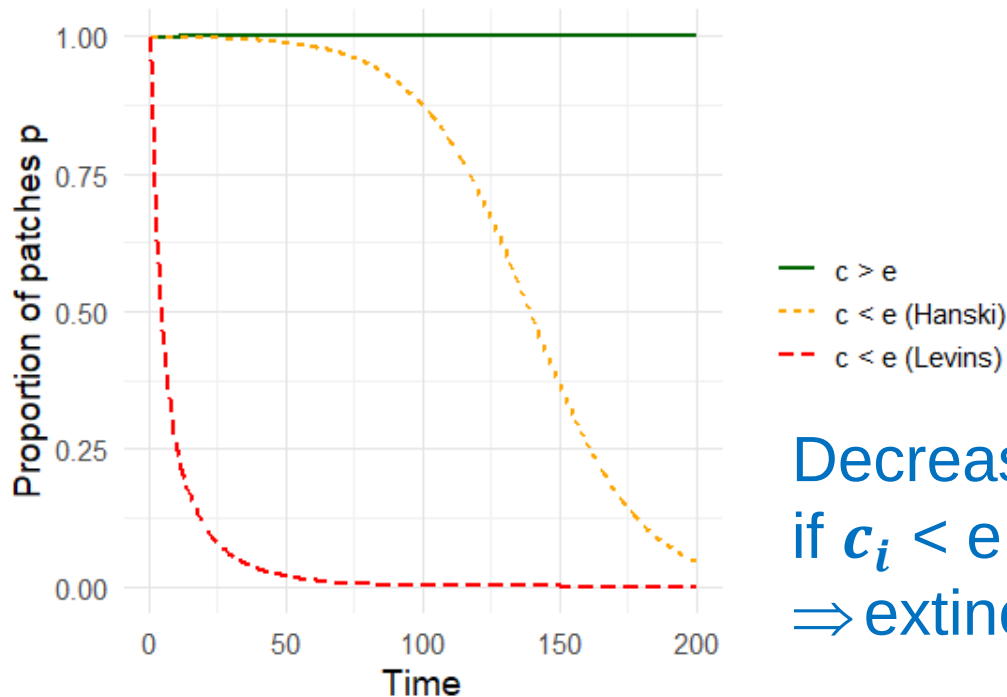
Phase diagram: Hanski model



Stability of equilibria

Question: What happen when patches shrink?

$$\frac{\partial \dot{p}}{\partial p} = (c_i - e)(1 - 2p)$$



Decreased c_i and increases e
if $c_i < e$, no stable positive occupancy
 $\Rightarrow$ extinction

Habitat destruction (Lande 1987, 1988; Kaareiva and Wennergren 1995):

$$\frac{dp}{dt} = c_i p(1 - D - p) - e p(1 - p)$$

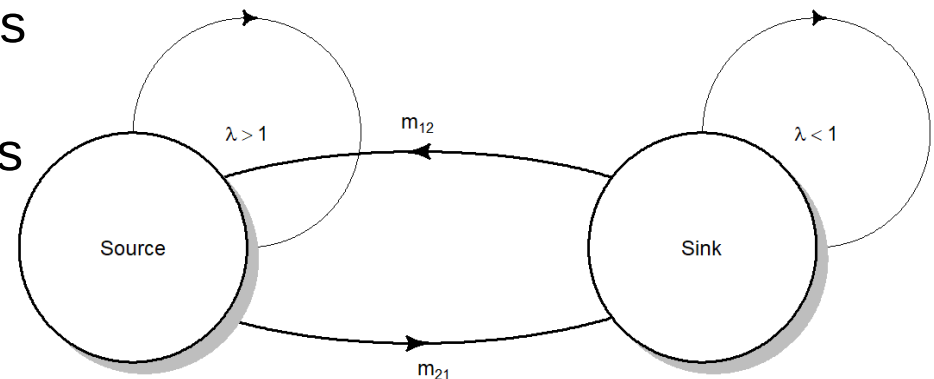
$\Rightarrow$ Test with $D = 0/0.2/0.5$

Metapopulation: Two-patches model

- Theory of source-sink dynamic: Dispersal from populations with positive per capita growth rates (sources) can rescue populations with negative per capita growth rates (sinks) from extinction (“rescue effect”).
- A case of two patches model:

$$\begin{aligned}\frac{dN_1}{dt} &= r_1 N_1 \left(1 - \frac{N_1}{K_1}\right) - e \left(\frac{N_1}{K_1}\right)^s N_1 + e N_2 \\ \frac{dN_2}{dt} &= r_2 N_2 + e \left(\frac{N_1}{K_1}\right)^s N_1 - e N_2\end{aligned}$$

N_1, N_2 = source and sink populations
 r_1 = intrinsic logistic growth rates
 r_2 = intrinsic exponential decay rates
 e = per capita emigration rate of population
 K_1 = carrying capacity
 t = time



Exercise 3 - Two-patch metapopulation model

Q3.1. Implement the two-patch metapopulation model in R using package simecol.

Model function:

$$dn1dt <- r1 * n1 * (1 - \alpha1 * n1) - e * n1 * (\alpha1 * n1)^s + e * n2$$

$$dn2dt <- r2 * n2 + e * n1 * (\alpha1 * n1)^s - e * n2$$

Q3.2. Why is there a decrease in the population for N2 at the beginning?

Q3.3. What is the effect of dispersal for N2 population?

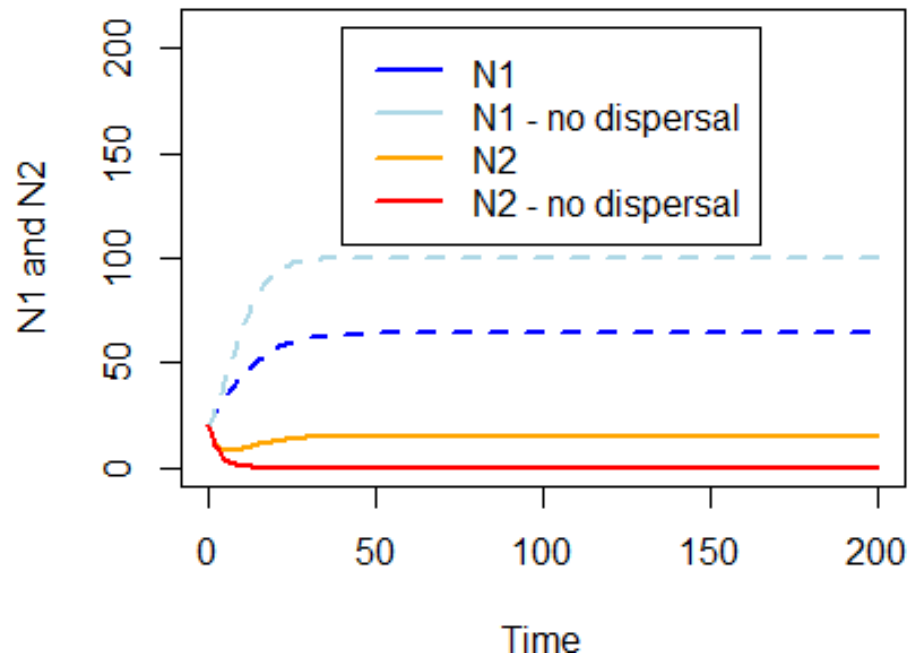
Test for $e = 0.0$ and 0.2 .

Exercise 3 - Two-patch metapopulation model

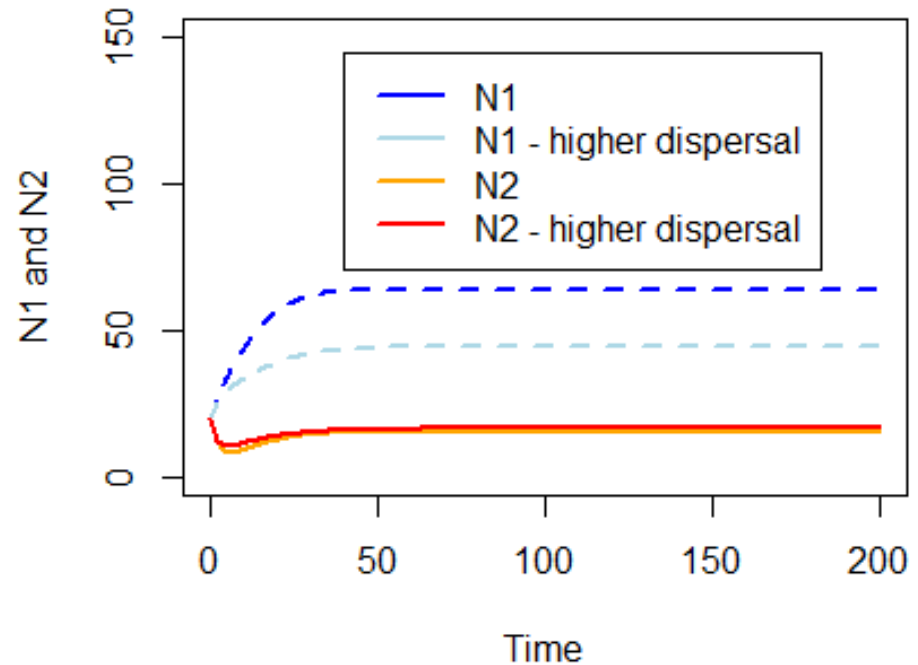
Q3.3. What is the effect of dispersal for N2 population?

Test for $e = 0.0$ and 0.2 .

Two-patch model: $e = 0.0$



Two-patch model: $e = 0.2$

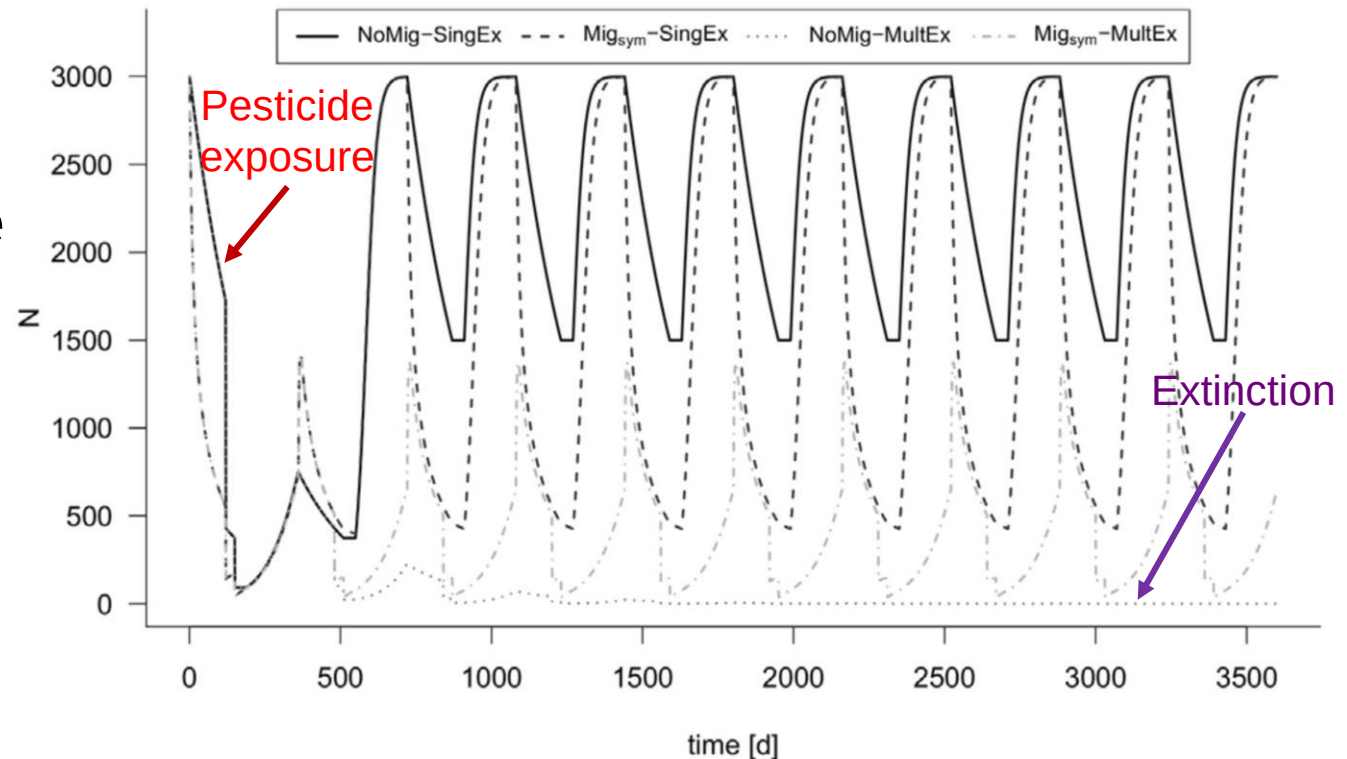


Extinction risks of source population

Ecological processes

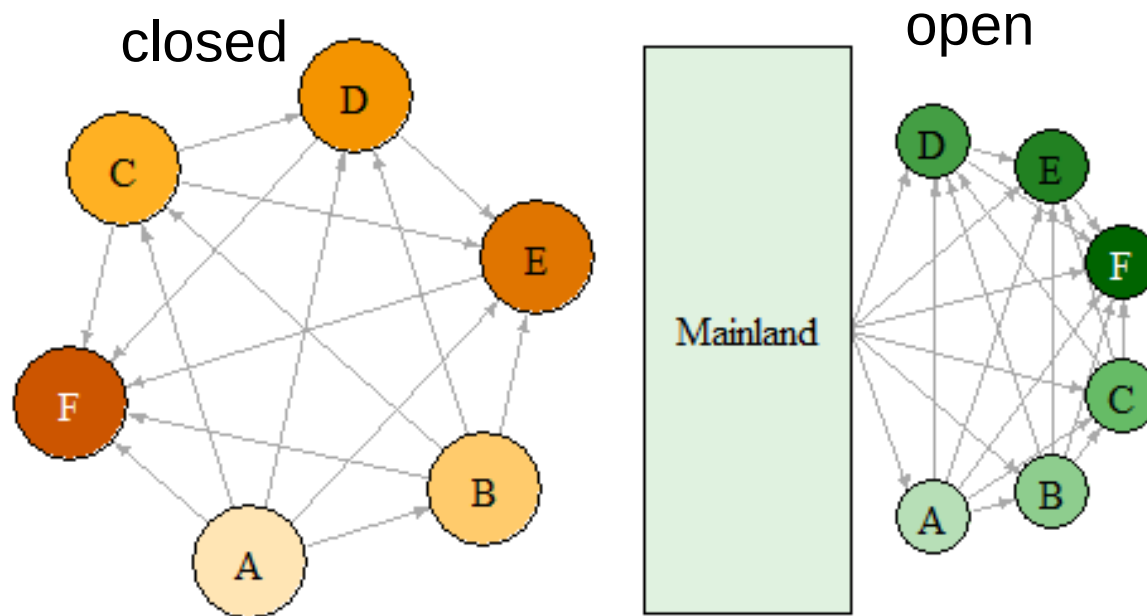
Dispersal influences temporal patterns

- Pesticide affect survival, reproduction and emergence of univoltine insects
- Extinction for scenarios with annual exposure and no migration
- Dispersal facilitates recovery, buffers against extinction in polluted patches, but carries effects to non-polluted patches.



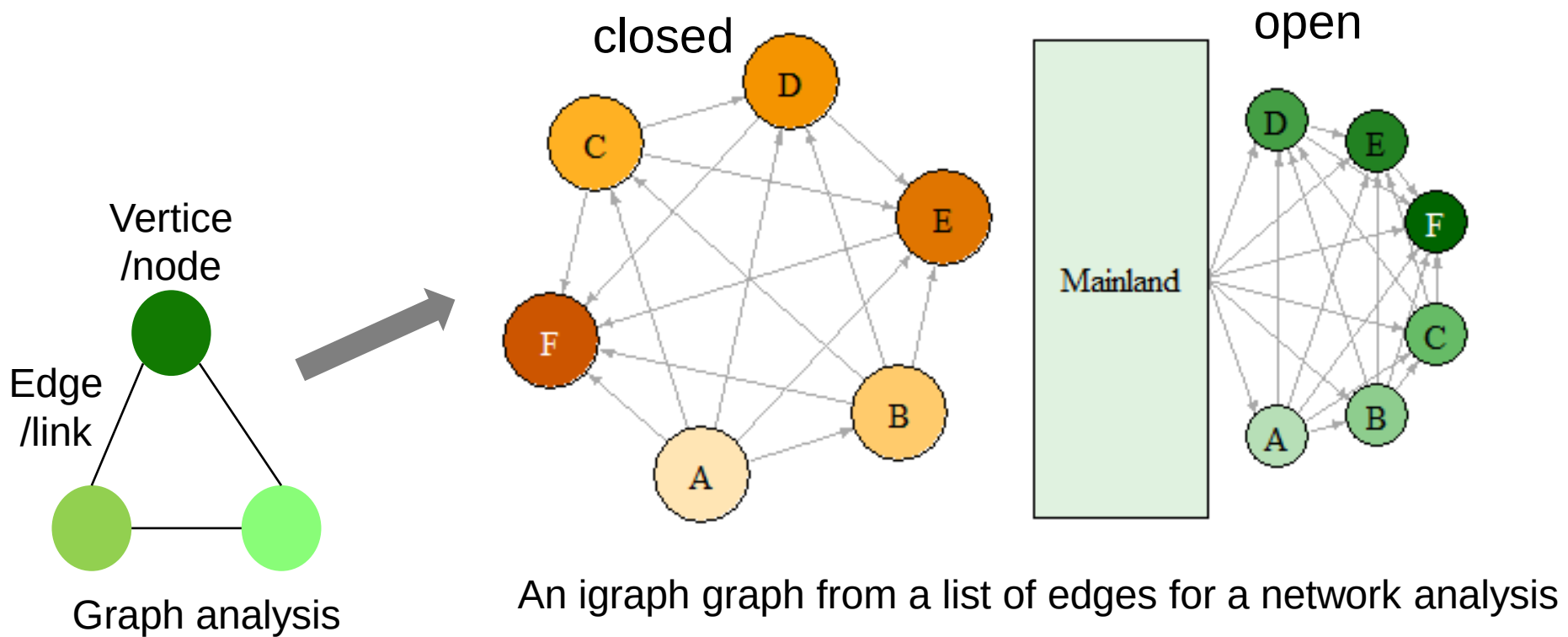
Metapopulation: Multiple patches

- Multiple patches are connected by dispersal (either closed or open)
- Either single population at different sites or metapopulation
- Modelling proportion of sites occupied



Metapopulation: Multiple patches

- **Metapopulation ecology:** focuses on patch-occupancy population dynamics and connectivity via species immigration/emigration, with landscapes treated as a network of connected patches.



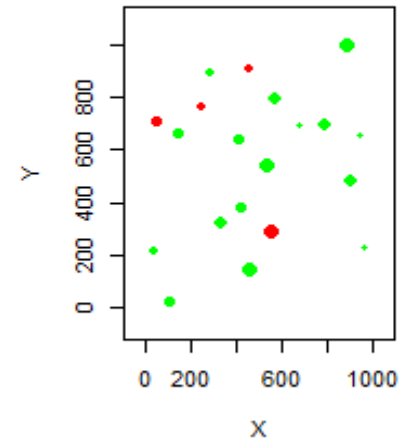
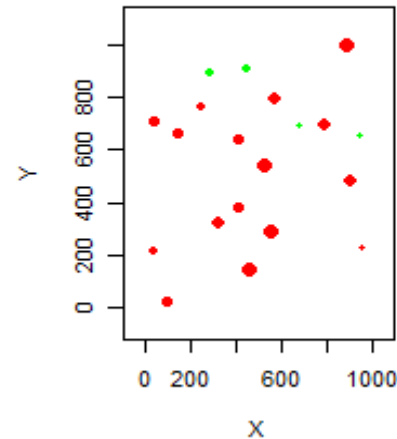
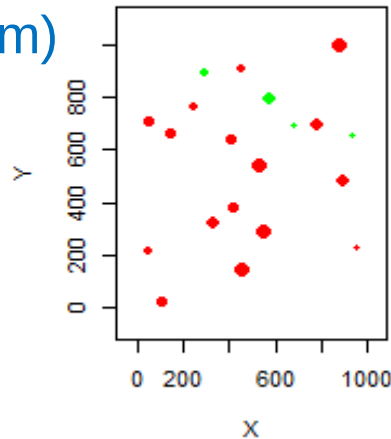
Exercise 4 - Multiple-patch metapopulation model

Implement metapopulation occupancy over a landscape in R.

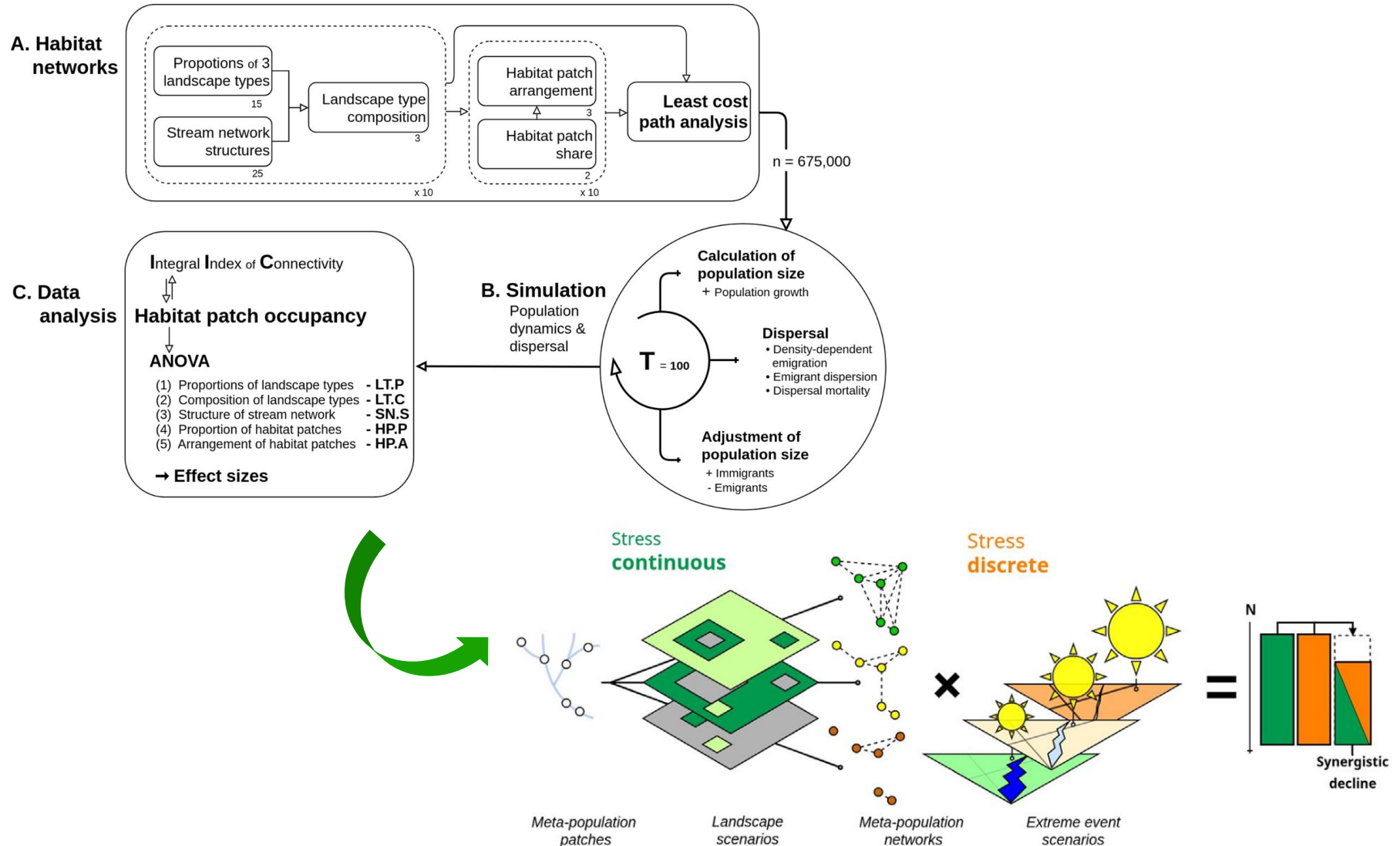
- Generate a random landscape as a graph
- Simulate metapopulation patch occupancy pattern
- Simulate metapopulation range expansion over the landscape

Question: How local extinction and dispersal affect spatial metapopulation pattern?

```
library(MetaLandSim)  
rland.graph()  
species.graph()  
simulate_graph()  
plotL.graph()
```



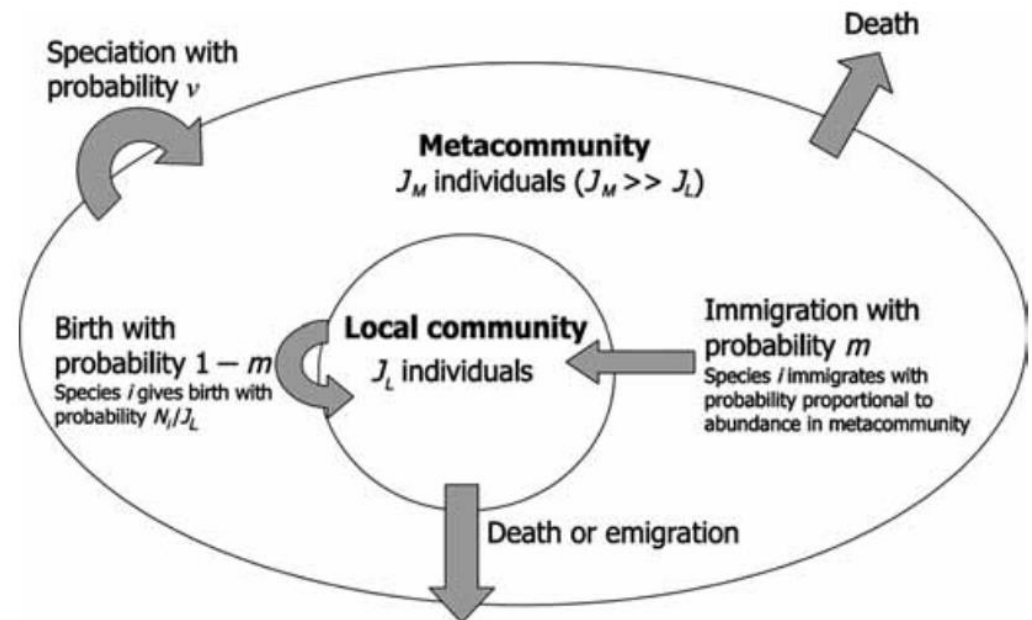
Metapopulation: Multiple patches



Streib et al. 2020 Ecol. Mod. 416: 108909;
Streib et al. 2022 STOTEN 814: 152602

Metacommunity models

- Metapopulation theory (Levins, 1969) expansion to include multi-species dynamics.
- Metacommunity: a network of local communities linked by dispersal of interacting species (Leibold, 2004).
- Theoretical framework:
 - Patch dynamics
 - Species-sorting
 - Mass effect
 - Neutral model

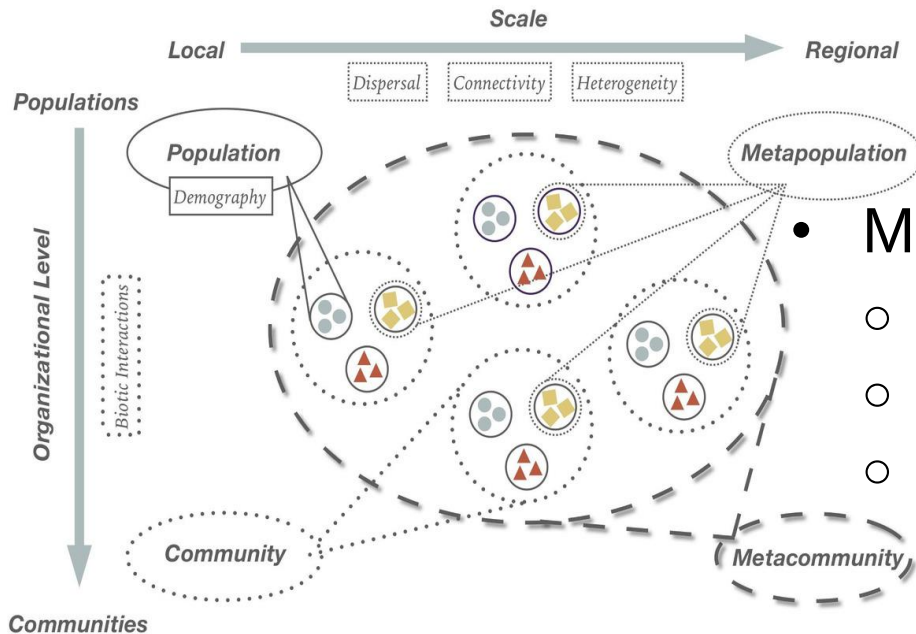
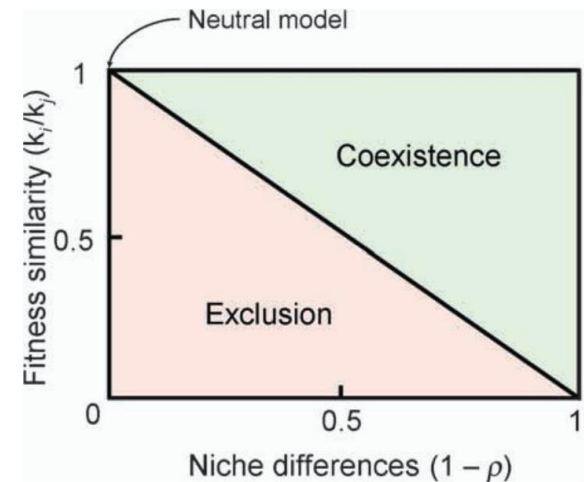


Leibold et al. 2004 Ecol. Letters 7: 601-613;

Cornell 2011 - In book: Encyclopedia of Theoretical Ecology

Metacommunity models

- Species coexistence theory:
 - Niche – fitness differences



- Metacommunity models:
 - Spatial/Landscape dynamics
 - Species dynamics
 - Interspecific species competition